

EDGAR COUNTY AP, IL, USA

WMO: 722172

Lat: 39.700N

Lon: 87.669W

Elev: 199

StdP: 98.95

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.1	-20.9	0.6	-16.6	-17.8	0.8	-13.9	13.3	0.6	12.0	1.4	2.3	310	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.9	23.9	31.7	23.9	30.4	23.3	26.4	30.6	25.4	29.5	24.5	28.5	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.9	29.4	24.1	19.5	27.9	23.2	18.5	26.8	83.6	30.7	79.0	29.4	75.2	28.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.4	DB	-21.1	34.5	5.2	1.8	-24.9	35.7	-27.9	36.8	-30.9	37.8	-34.7	39.1	(4)
(5)			WB	-21.4	27.9	5.1	0.9	-25.1	28.6	-28.1	29.1	-31.0	29.5	-34.7	30.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-2.3	-0.5	5.8	12.0	18.0	22.3	23.1	22.4	19.5	12.9	5.9	0.6	(6)	
	DBStd	10.45	6.87	6.61	6.32	5.28	4.94	3.17	3.08	3.05	3.92	5.26	5.35	5.91	(7)	
	HDD10.0	1354	383	297	164	41	3	0	0	0	0	27	143	295	(8)	
	HDD18.3	2975	639	527	392	198	69	6	2	4	31	184	373	549	(9)	
	CDD10.0	1976	2	4	33	99	253	370	405	383	286	118	21	4	(10)	
	CDD18.3	555	0	0	2	7	60	126	148	130	66	16	0	0	(11)	
	CDH23.3	5202	0	0	17	88	558	1127	1379	1201	687	144	1	0	(12)	
	CDH26.7	1750	0	0	2	10	169	386	495	408	239	41	0	0	(13)	
(14) Wind	WSAvg	3.3	4.4	4.0	4.1	4.3	3.5	2.6	1.8	1.7	1.9	3.2	3.8	3.9	(14)	
(15) Precipitation	PrecAvg	1036	61	54	82	101	111	99	112	98	79	71	88	78	(15)	
	PrecMax	1452	202	139	190	228	318	323	246	209	242	255	266	231	(16)	
	PrecMin	777	2	10	18	8	19	12	14	11	4	12	11	15	(17)	
	PrecStd	169	44	32	42	52	67	60	65	54	60	46	55	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	18.5	25.9	27.9	32.0	33.5	36.5	33.9	33.8	31.0	22.4	17.3	(19)	
		MCWB	14.2	14.3	18.3	17.1	21.3	23.2	24.6	24.4	23.0	22.0	15.1	15.0	(20)	
	2%	DB	12.8	15.1	22.0	26.0	30.4	32.1	32.6	32.1	32.0	27.3	19.5	14.2	(21)	
		MCWB	11.3	13.3	16.0	17.0	21.3	23.7	25.1	24.4	22.5	19.3	14.9	12.7	(22)	
	5%	DB	10.2	12.0	18.4	23.7	28.7	30.7	31.1	30.5	29.5	24.7	16.8	11.6	(23)	
		MCWB	8.8	9.9	14.3	16.1	20.3	23.3	25.1	24.0	21.7	17.7	13.4	10.4	(24)	
	10%	DB	7.0	9.1	16.0	21.3	26.9	28.9	29.4	29.1	27.4	22.2	14.0	8.8	(25)	
		MCWB	5.6	7.1	12.7	15.1	19.7	22.1	23.9	23.2	19.6	16.9	10.6	7.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.7	18.8	20.0	24.1	26.5	27.9	27.5	25.3	22.8	18.3	15.6	(27)	
		MCDB	15.9	17.6	24.9	24.7	29.1	31.0	31.9	31.4	30.9	28.8	20.8	16.9	(28)	
	2%	WB	12.0	13.2	16.9	18.4	22.6	25.2	26.7	26.2	23.8	21.2	15.8	12.9	(29)	
		MCDB	12.7	14.7	20.9	22.7	28.0	29.8	30.7	30.4	28.8	25.4	17.8	13.8	(30)	
	5%	WB	9.0	10.5	15.3	17.3	21.6	24.2	25.7	25.1	22.7	19.7	13.7	10.2	(31)	
		MCDB	10.0	12.1	18.1	21.7	26.5	28.7	29.5	29.1	27.3	22.4	16.4	11.2	(32)	
	10%	WB	5.5	7.0	13.0	16.0	20.6	23.3	24.7	24.1	21.6	17.8	11.5	7.5	(33)	
		MCDB	6.9	8.8	15.5	20.2	25.2	27.3	28.5	27.5	25.6	20.8	13.8	8.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.0	10.0	11.7	11.2	11.0	11.0	11.9	13.4	11.8	9.9	7.6	(35)	
		MCDBR	11.1	12.4	13.3	14.9	13.2	12.6	12.2	12.8	15.1	14.1	12.8	10.4	(36)	
	5% WB	MCWBR	10.0	10.4	8.6	7.9	5.8	5.9	6.0	6.3	6.5	7.3	8.8	9.3	(37)	
		MCWBR	10.8	12.1	12.0	12.3	11.5	10.3	10.8	10.8	12.3	11.7	11.2	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.316	0.356	0.405	0.441	0.445	0.464	0.473	0.427	0.364	0.317	0.299	(40)	
		taud	2.440	2.397	2.362	2.227	2.178	2.208	2.146	2.109	2.218	2.396	2.482	2.473	(41)	
	Ebn at Noon		866	904	898	870	842	837	817	797	811	836	844	844	(42)	
		Edh at Noon	80	97	112	136	147	142	150	152	128	96	77	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.78	2.63	3.67	4.76	5.43	6.23	6.09	5.51	4.61	3.22	2.15	1.51	(44)	
	RadStd		0.20	0.28	0.30	0.33	0.48	0.38	0.36	0.34	0.33	0.32	0.22	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page